

Byzantine Robustness Comparison up to $f = 10$: CNN vs. SNN Encodings (Complete Sweeps)

ByzFL SNN Benchmark

June 24, 2026

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1 Introduction

This document presents comparison grids (2x2 layout) of Byzantine robustness results for CNN and SNN architectures under different data encoding schemes, extended up to $f = 10$ Byzantine clients. The configurations compared are:

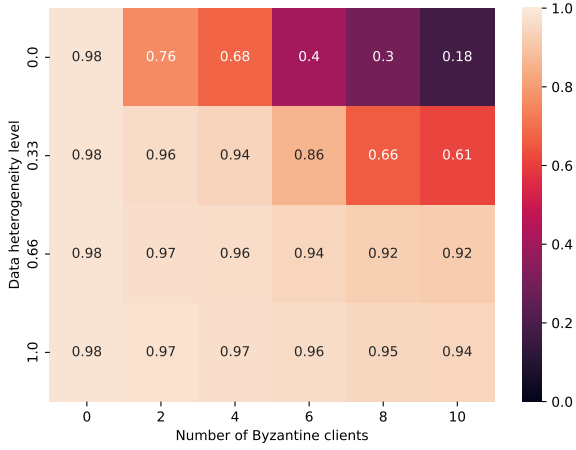
1. **CNN**: Convolutional Neural Network on MNIST ('cnn_complete_plots_extended_f10')
2. **SNN Rate**: Spiking Neural Network with Rate encoding on MNIST ('snn_complete_plots_rate_extended_f10')
3. **SNN Direct**: Spiking Neural Network with Direct encoding on MNIST ('snn_complete_plots_direct_extended_f10')
4. **SNN NMNIST**: Spiking Neural Network on neuromorphic NMNIST ('snn_complete_plots_nmnist_extended_f10')

Each grid illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 6, 8, 10]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$).

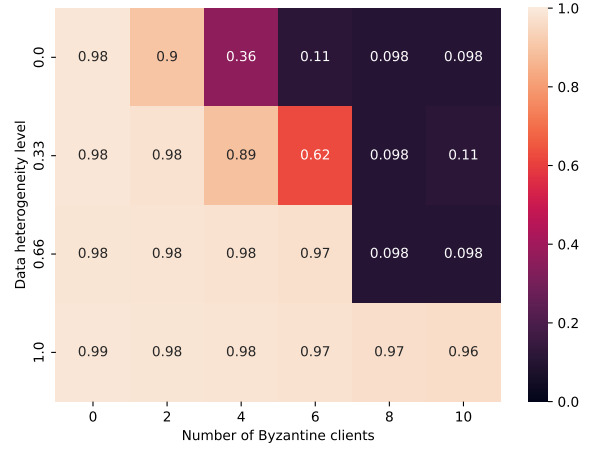
2 Best Overall (Worst-Case Across All Attacks)

This section compares the performance under the worst-case attack scenario (minimizing the maximum damage over all attacks), showing both the dynamically selected best aggregator (using both validation-based best step and max test accuracy metrics) and each of the fixed aggregators.

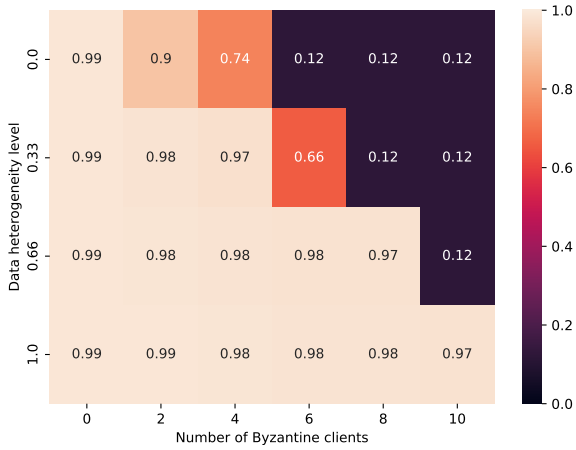
2.1 Best Aggregator Choice



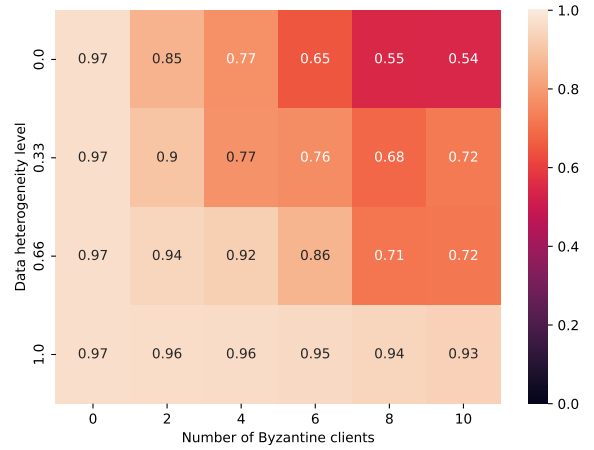
(a) CNN (Validation-Based Best Step)



(b) SNN Rate (Validation-Based Best Step)

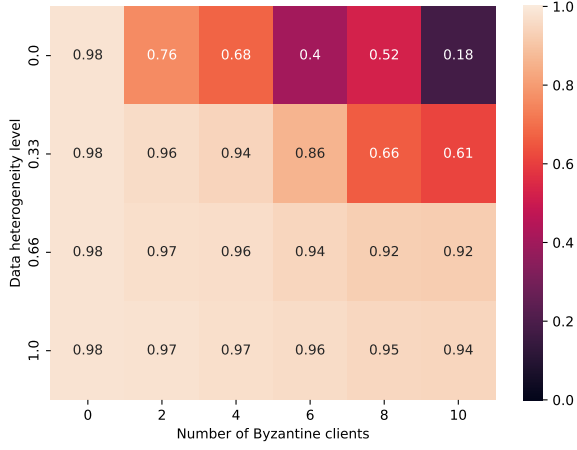


(c) SNN Direct (Validation-Based Best Step)

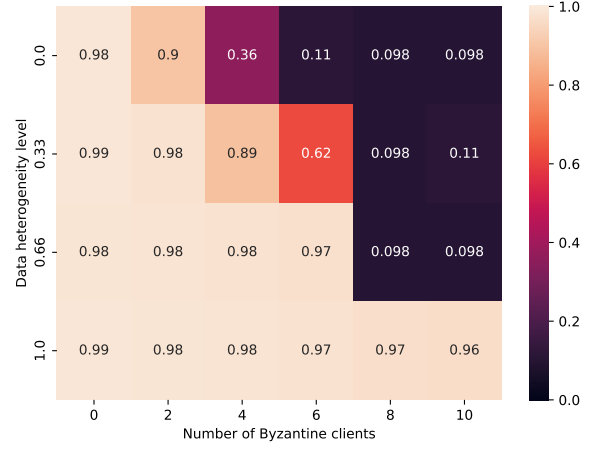


(d) SNN MNIST (Validation-Based Best Step)

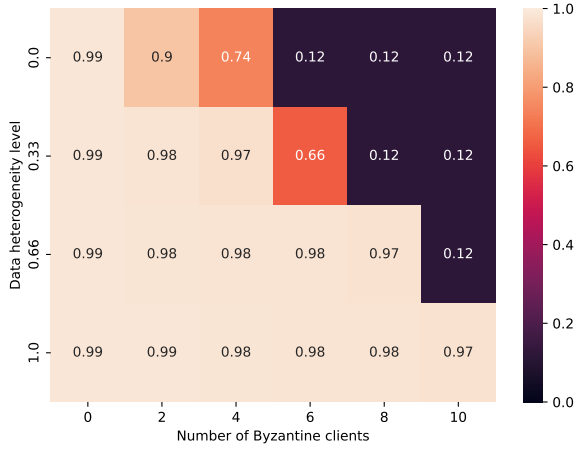
Figure 1: Worst-case test accuracy comparison using the best overall aggregator selection evaluated at the validation-based best step.



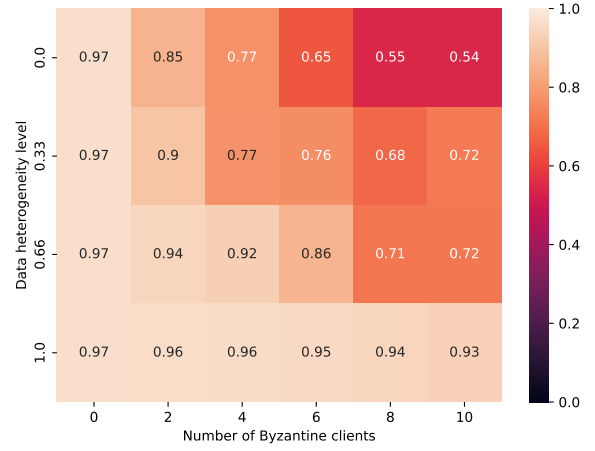
(a) CNN (Max Test Accuracy)



(b) SNN Rate (Max Test Accuracy)



(c) SNN Direct (Max Test Accuracy)



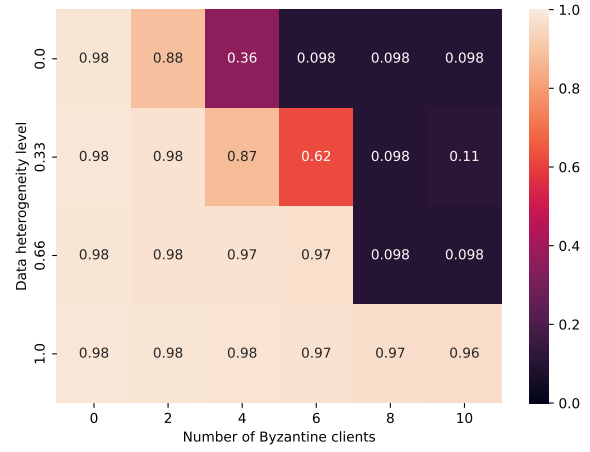
(d) SNN NMNIST (Max Test Accuracy)

Figure 2: Worst-case test accuracy comparison using the best overall aggregator selection evaluated at the max test accuracy.

2.2 Centered Clipping



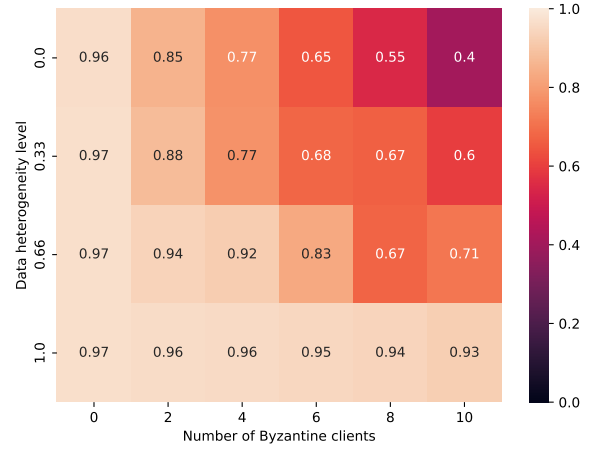
(a) CNN



(b) SNN Rate



(c) SNN Direct



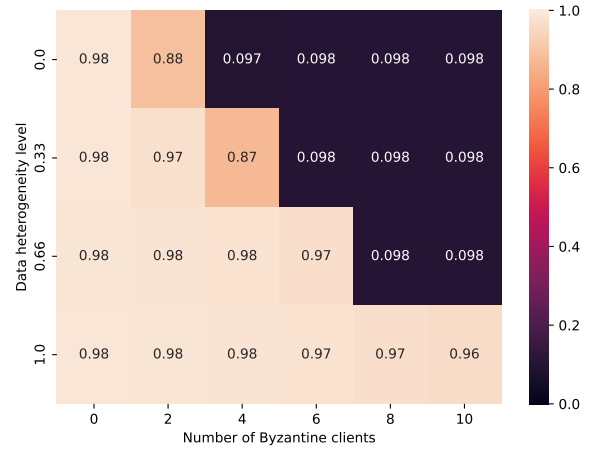
(d) SNN MNIST

Figure 3: Worst-case test accuracy comparison using Centered Clipping.

2.3 Geometric Median



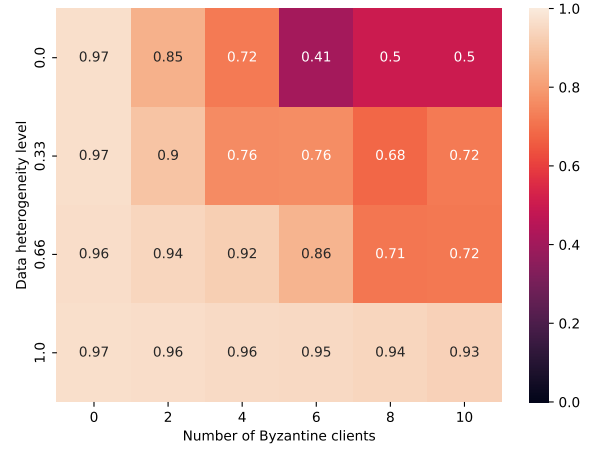
(a) CNN



(b) SNN Rate



(c) SNN Direct



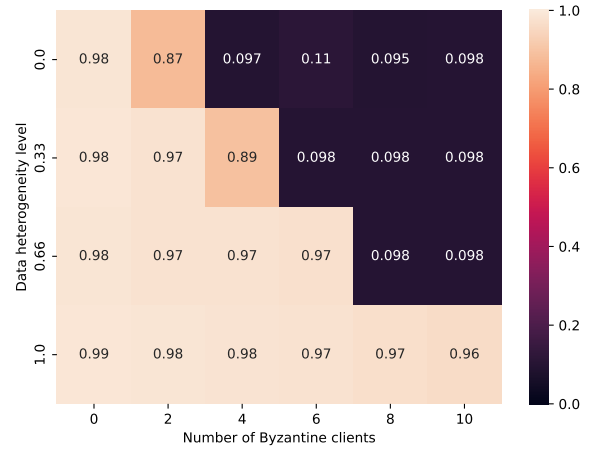
(d) SNN NMNIST

Figure 4: Worst-case test accuracy comparison using Geometric Median.

2.4 Multi-Krum



(a) CNN



(b) SNN Rate



(c) SNN Direct



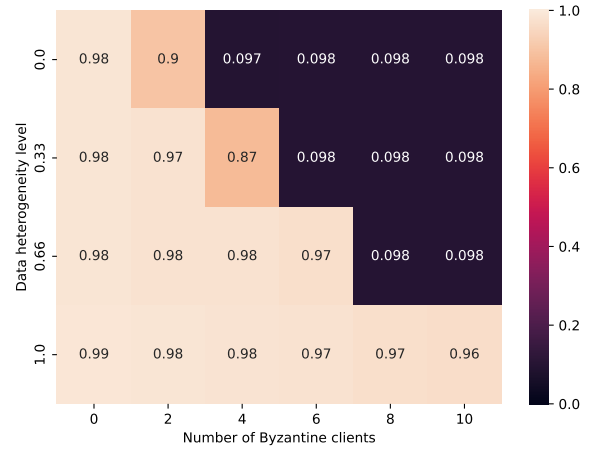
(d) SNN NMNIST

Figure 5: Worst-case test accuracy comparison using Multi-Krum.

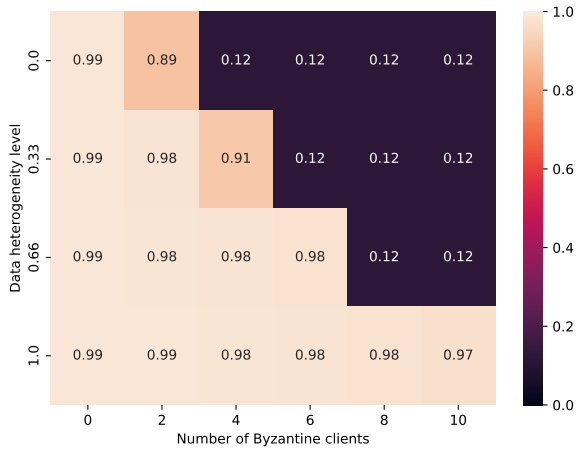
2.5 Trimmed Mean



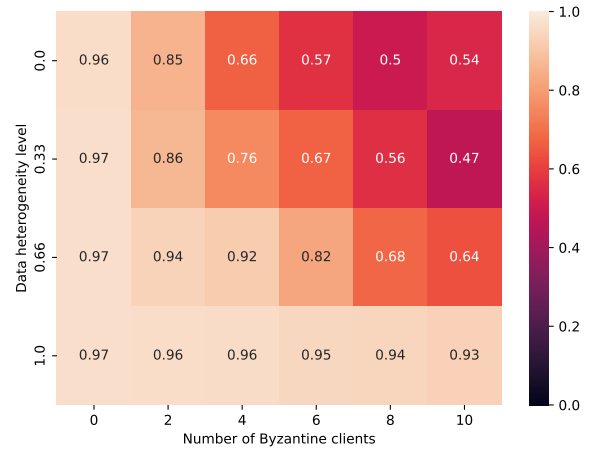
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 6: Worst-case test accuracy comparison using Trimmed Mean.

3 Fixed Attack: Optimal A Little Is Enough (ALIE)

This section compares accuracy under the Optimal ALIE Byzantine attack.

3.1 Best Aggregator Choice

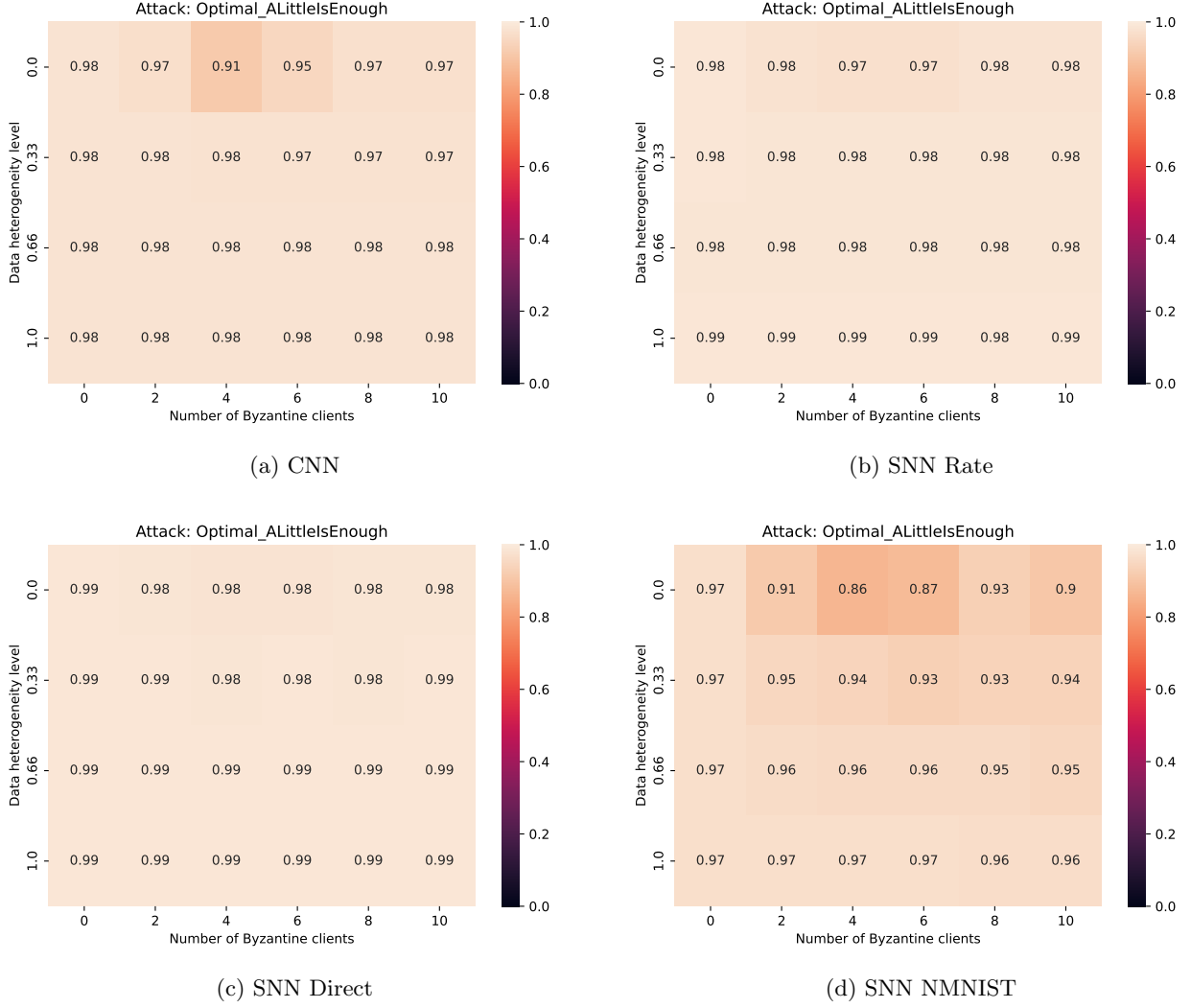


Figure 7: Accuracy comparison under Optimal ALIE attack using the best aggregator selection.

3.2 Centered Clipping



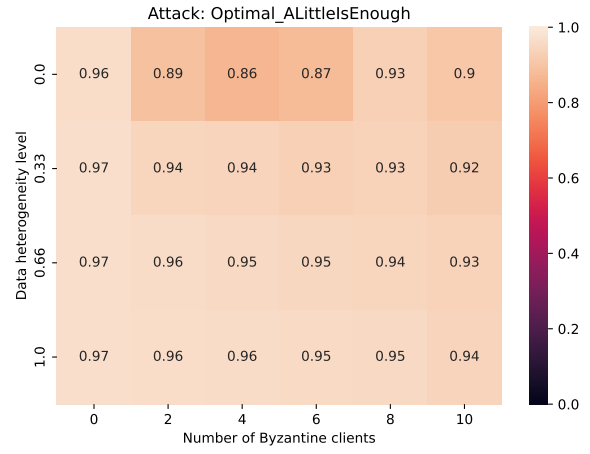
(a) CNN



(b) SNN Rate



(c) SNN Direct



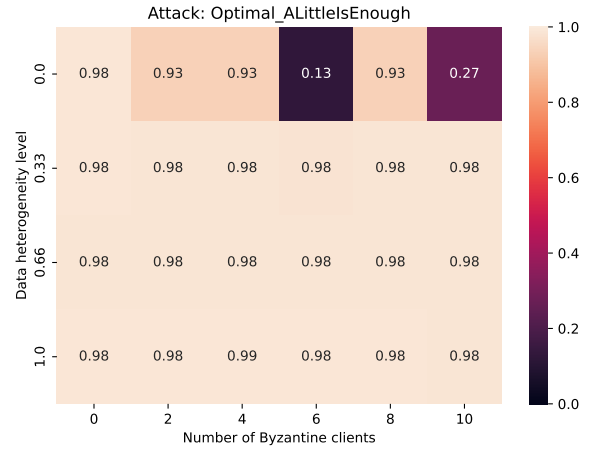
(d) SNN MNIST

Figure 8: Accuracy comparison under Optimal ALIE attack using Centered Clipping.

3.3 Geometric Median



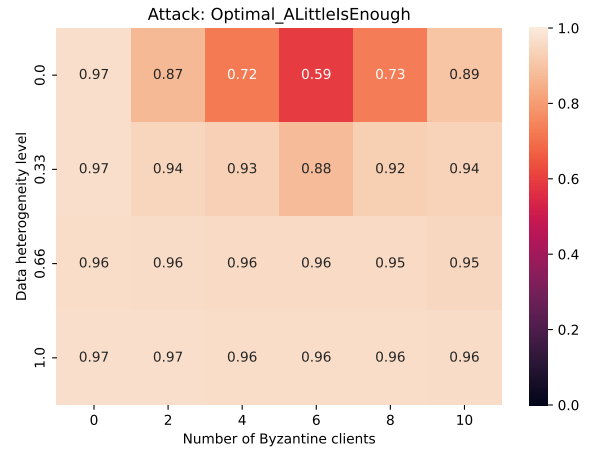
(a) CNN



(b) SNN Rate



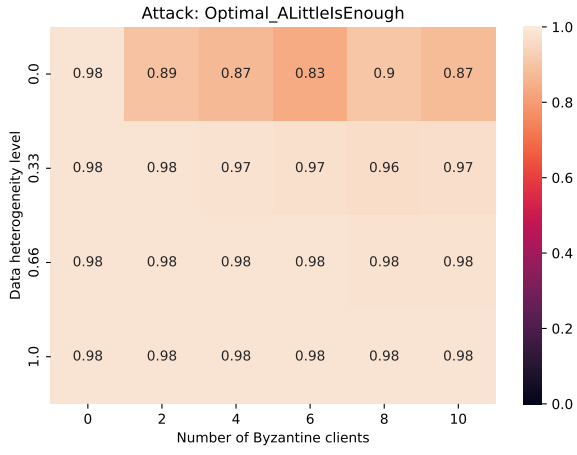
(c) SNN Direct



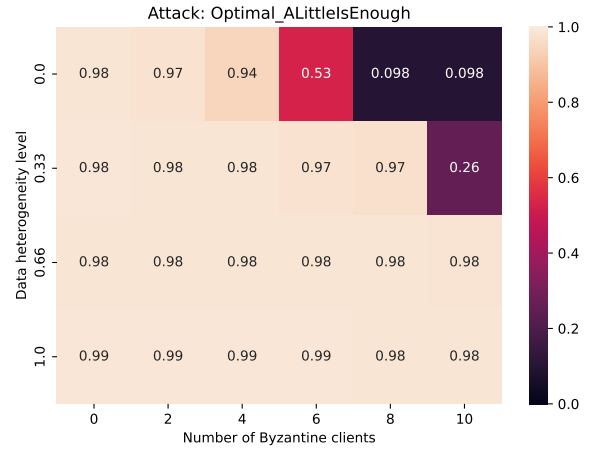
(d) SNN MNIST

Figure 9: Accuracy comparison under Optimal ALIE attack using Geometric Median.

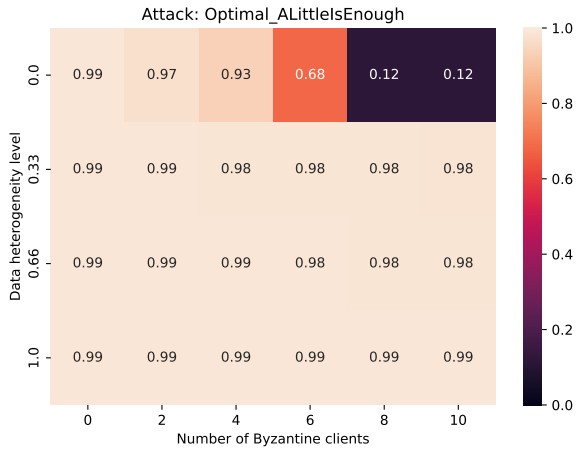
3.4 Multi-Krum



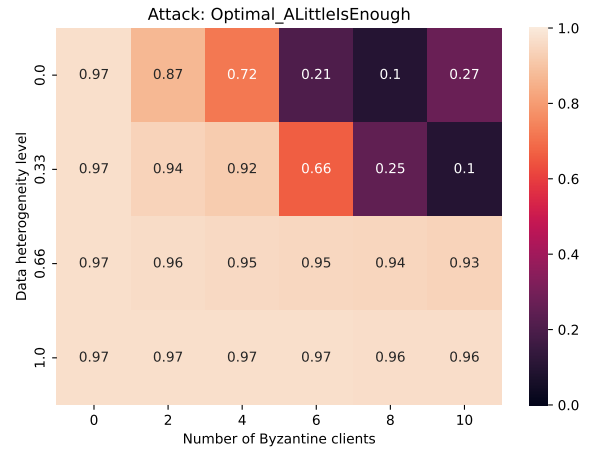
(a) CNN



(b) SNN Rate



(c) SNN Direct



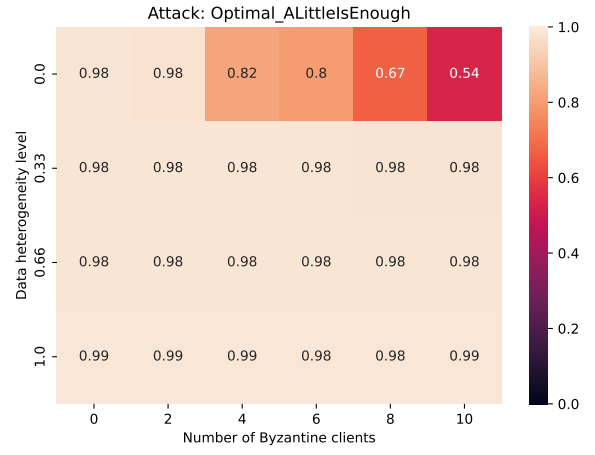
(d) SNN MNIST

Figure 10: Accuracy comparison under Optimal ALIE attack using Multi-Krum.

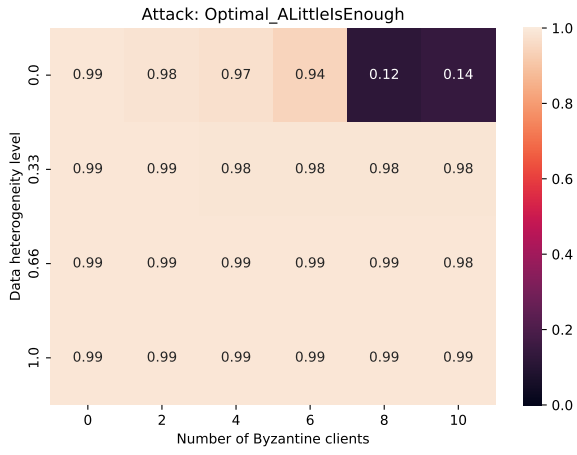
3.5 Trimmed Mean



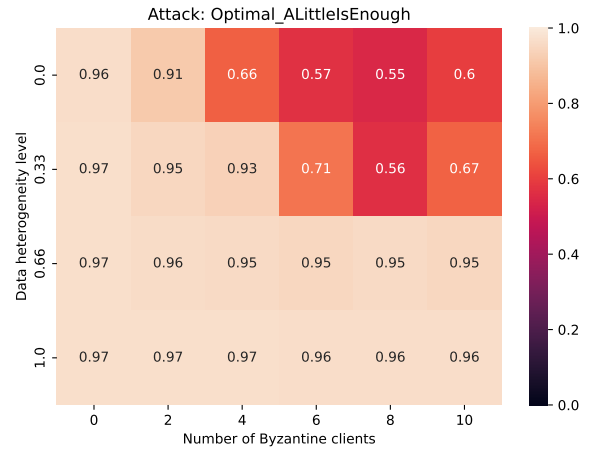
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 11: Accuracy comparison under Optimal ALIE attack using Trimmed Mean.

4 Fixed Attack: Optimal Inner Product Manipulation (IPM)

This section compares accuracy under the Optimal IPM Byzantine attack.

4.1 Best Aggregator Choice

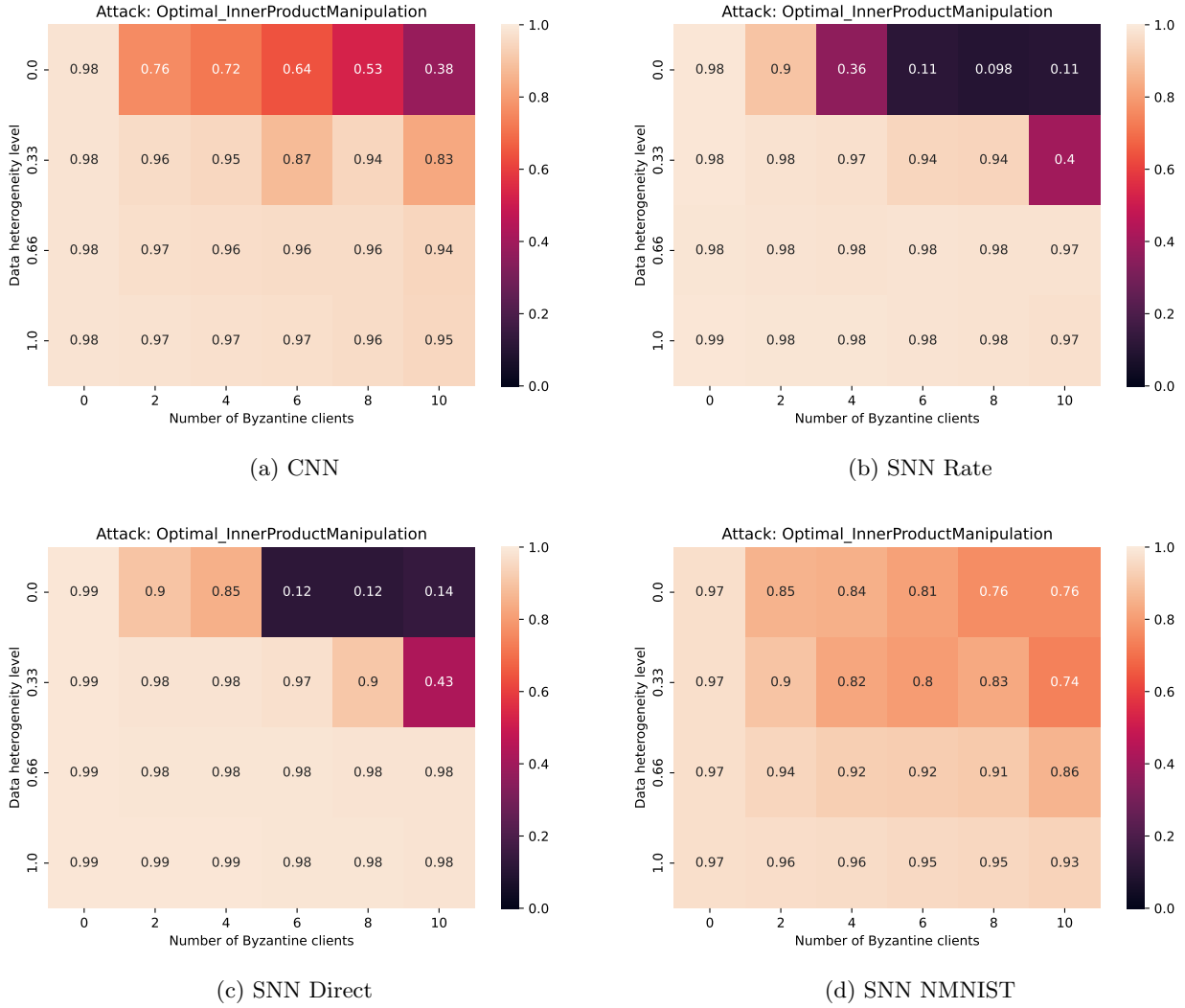
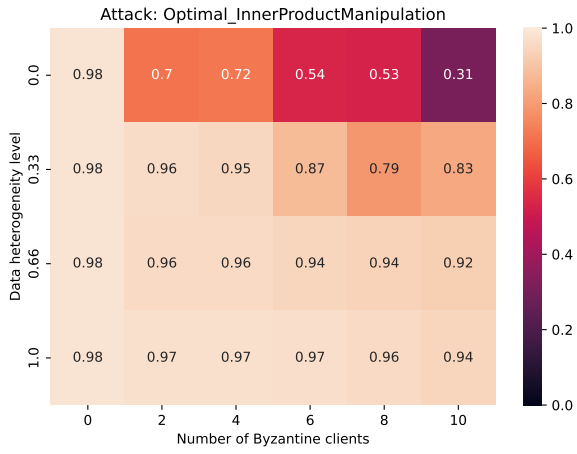
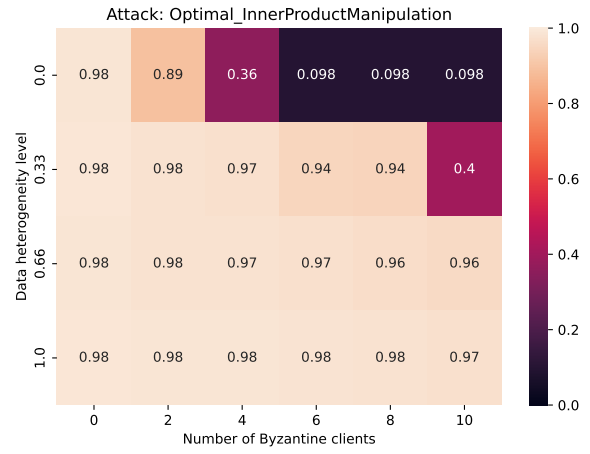


Figure 12: Accuracy comparison under Optimal IPM attack using the best aggregator selection.

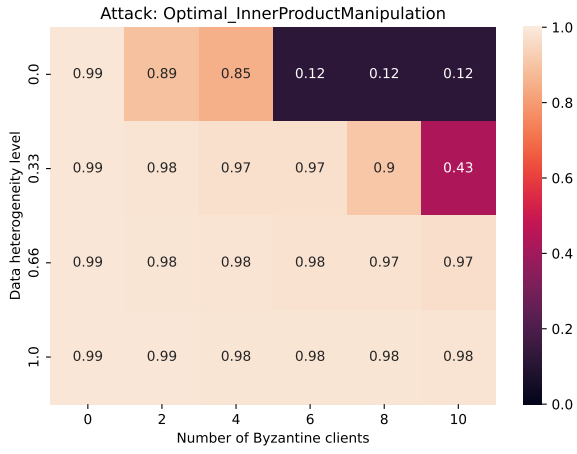
4.2 Centered Clipping



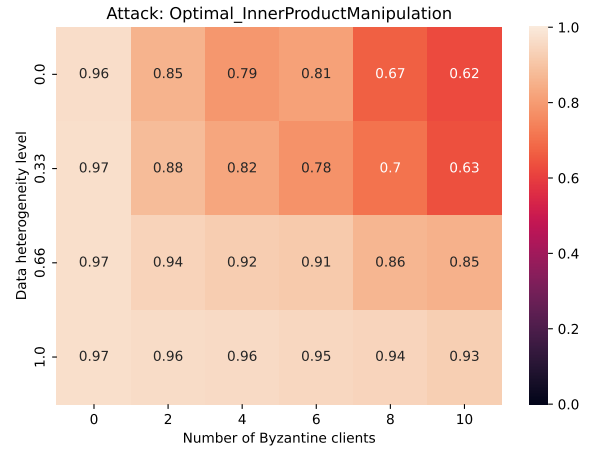
(a) CNN



(b) SNN Rate



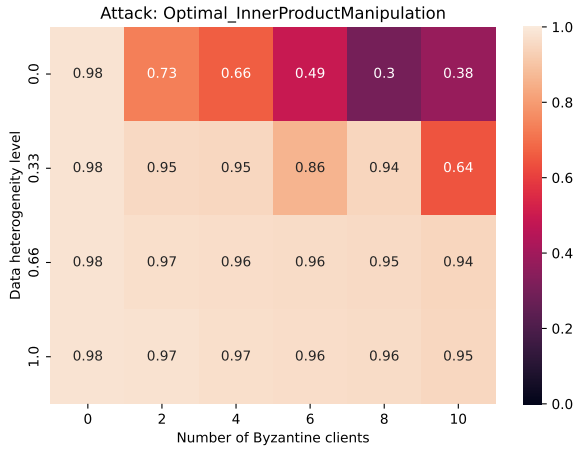
(c) SNN Direct



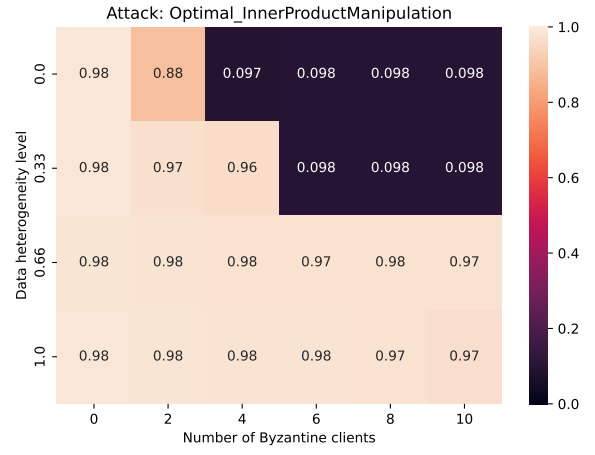
(d) SNN NMNIST

Figure 13: Accuracy comparison under Optimal IPM attack using Centered Clipping.

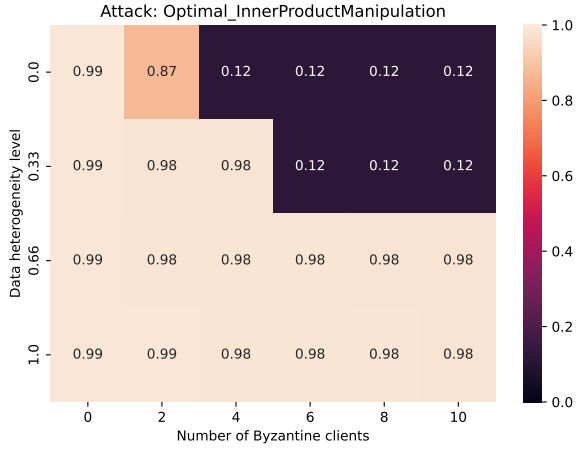
4.3 Geometric Median



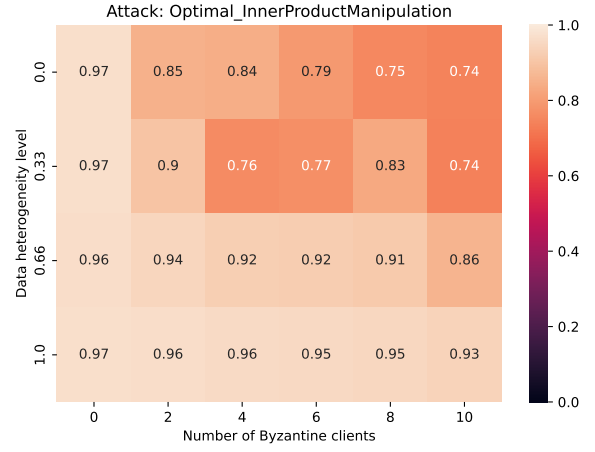
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 14: Accuracy comparison under Optimal IPM attack using Geometric Median.

4.4 Multi-Krum

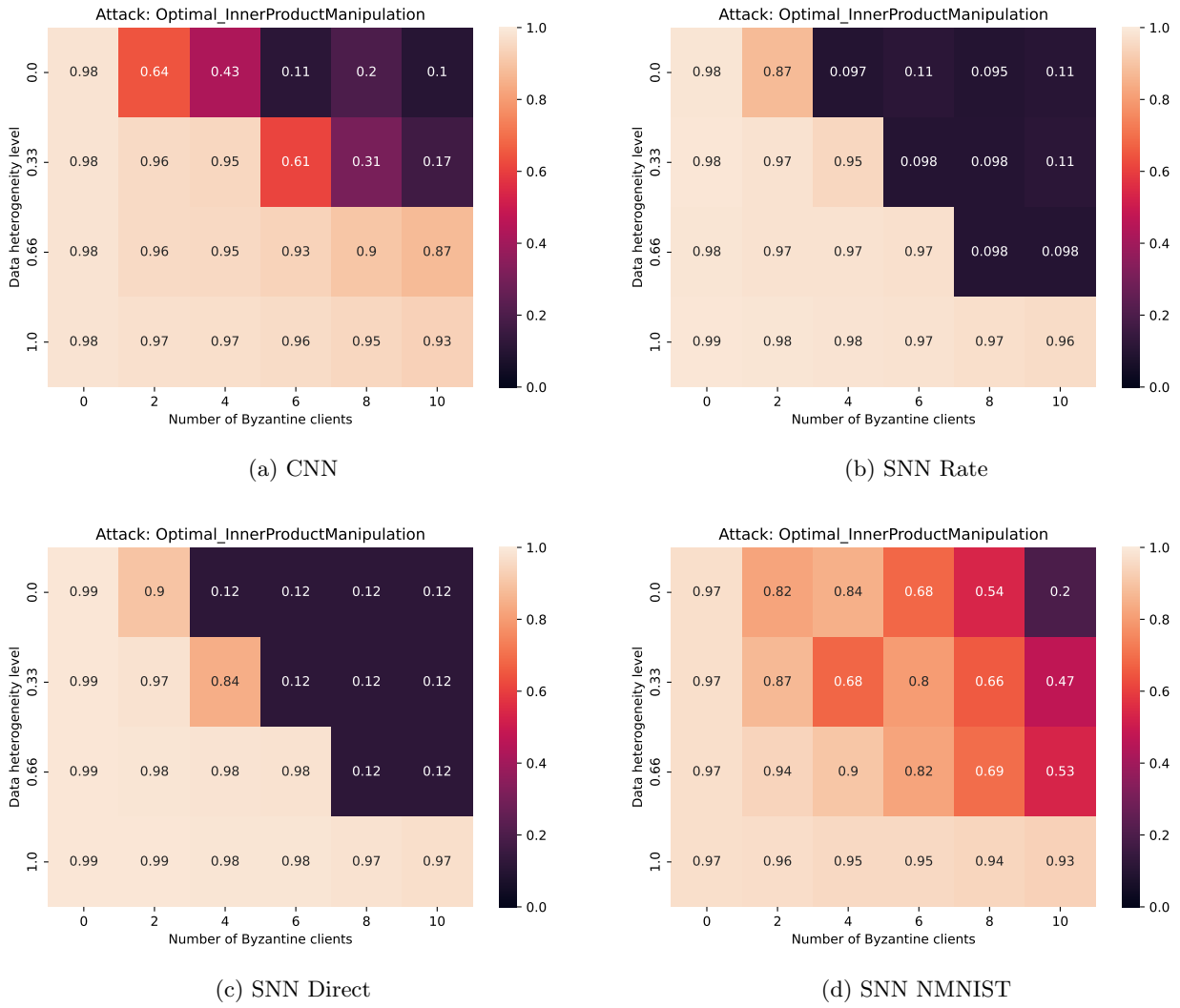
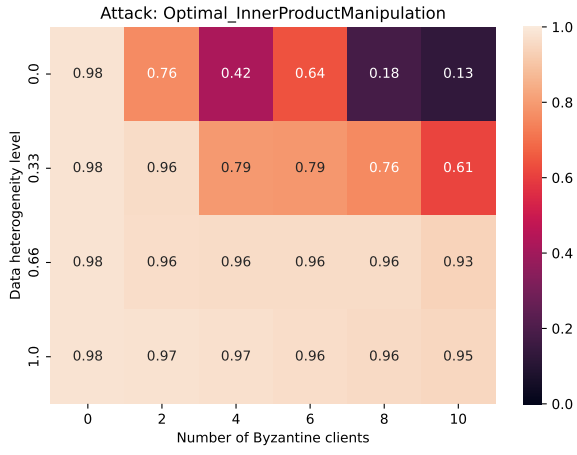
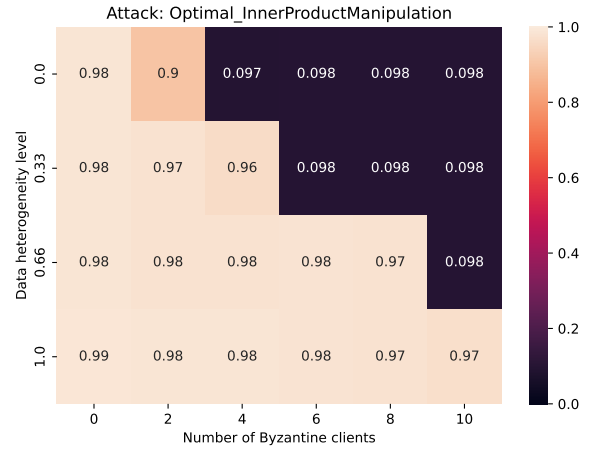


Figure 15: Accuracy comparison under Optimal IPM attack using Multi-Krum.

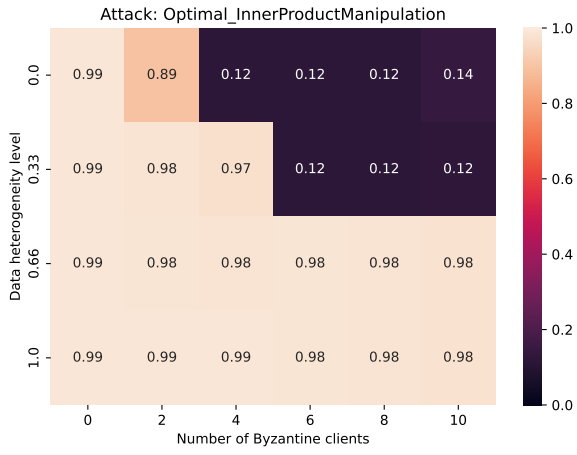
4.5 Trimmed Mean



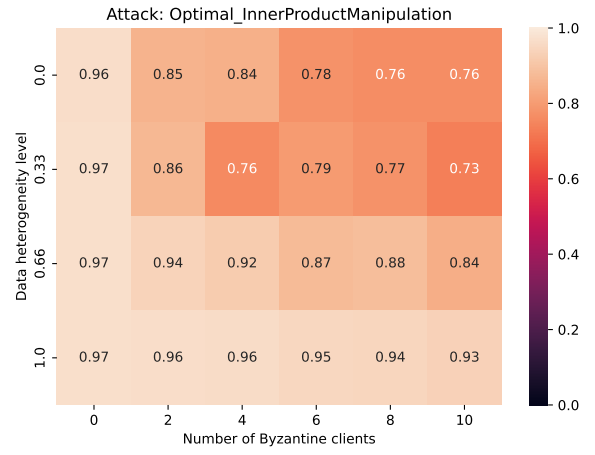
(a) CNN



(b) SNN Rate



(c) SNN Direct



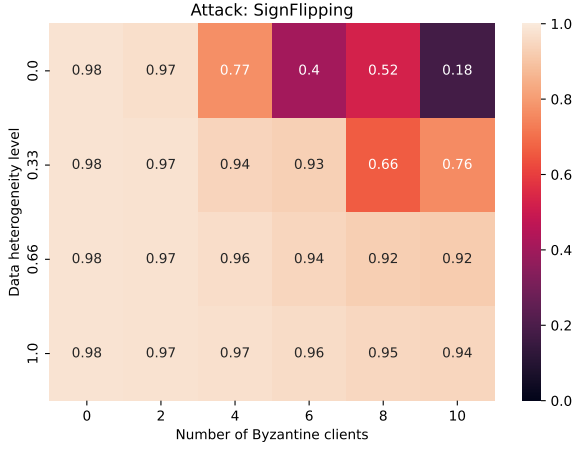
(d) SNN NMNIST

Figure 16: Accuracy comparison under Optimal IPM attack using Trimmed Mean.

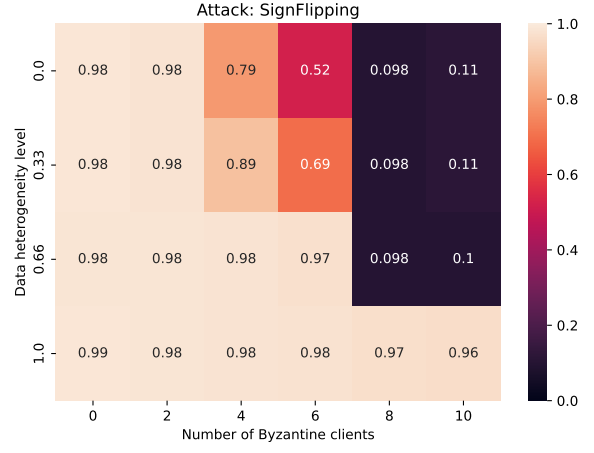
5 Fixed Attack: Sign Flipping

This section compares accuracy under the Sign Flipping Byzantine attack.

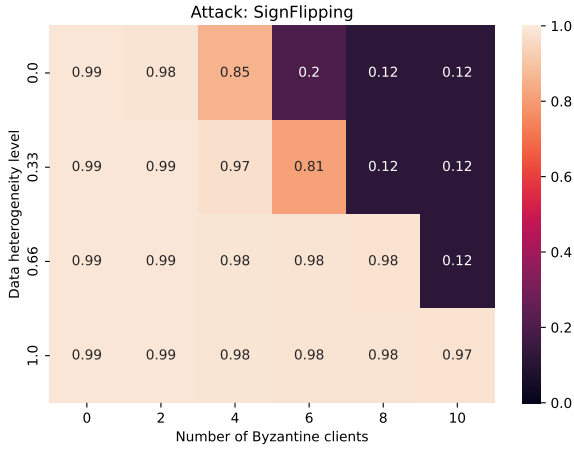
5.1 Best Aggregator Choice



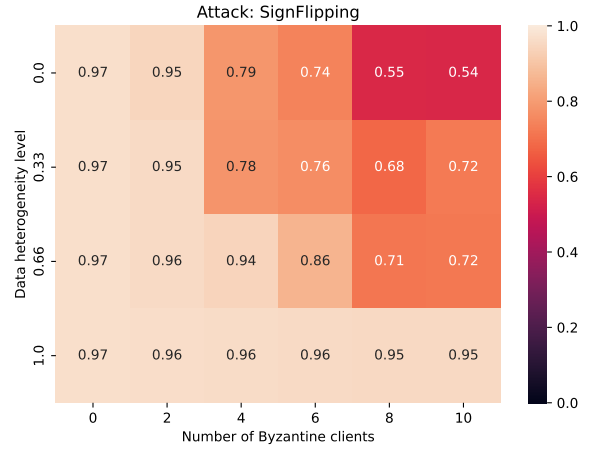
(a) CNN



(b) SNN Rate



(c) SNN Direct



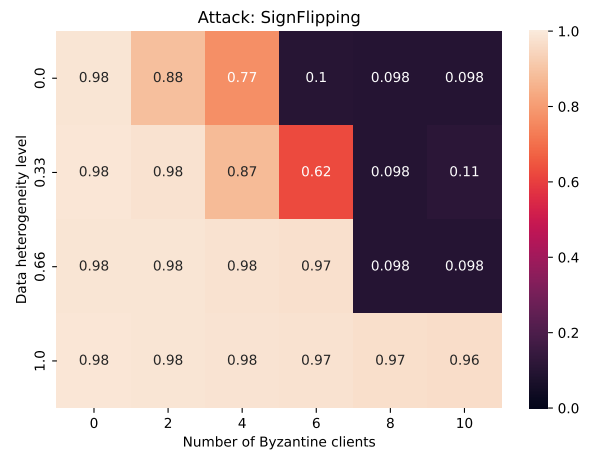
(d) SNN MNIST

Figure 17: Accuracy comparison under Sign Flipping attack using the best aggregator selection.

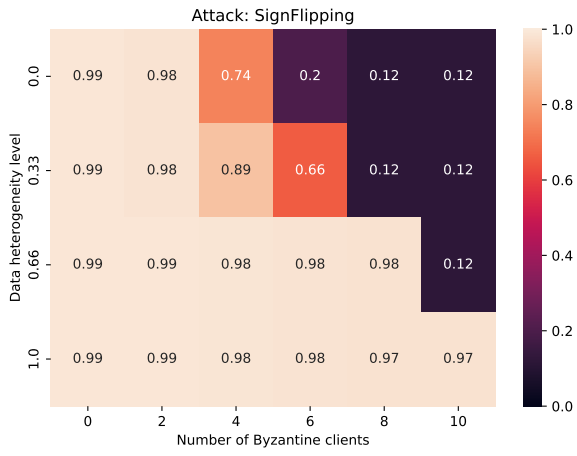
5.2 Centered Clipping



(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 18: Accuracy comparison under Sign Flipping attack using Centered Clipping.

5.3 Geometric Median

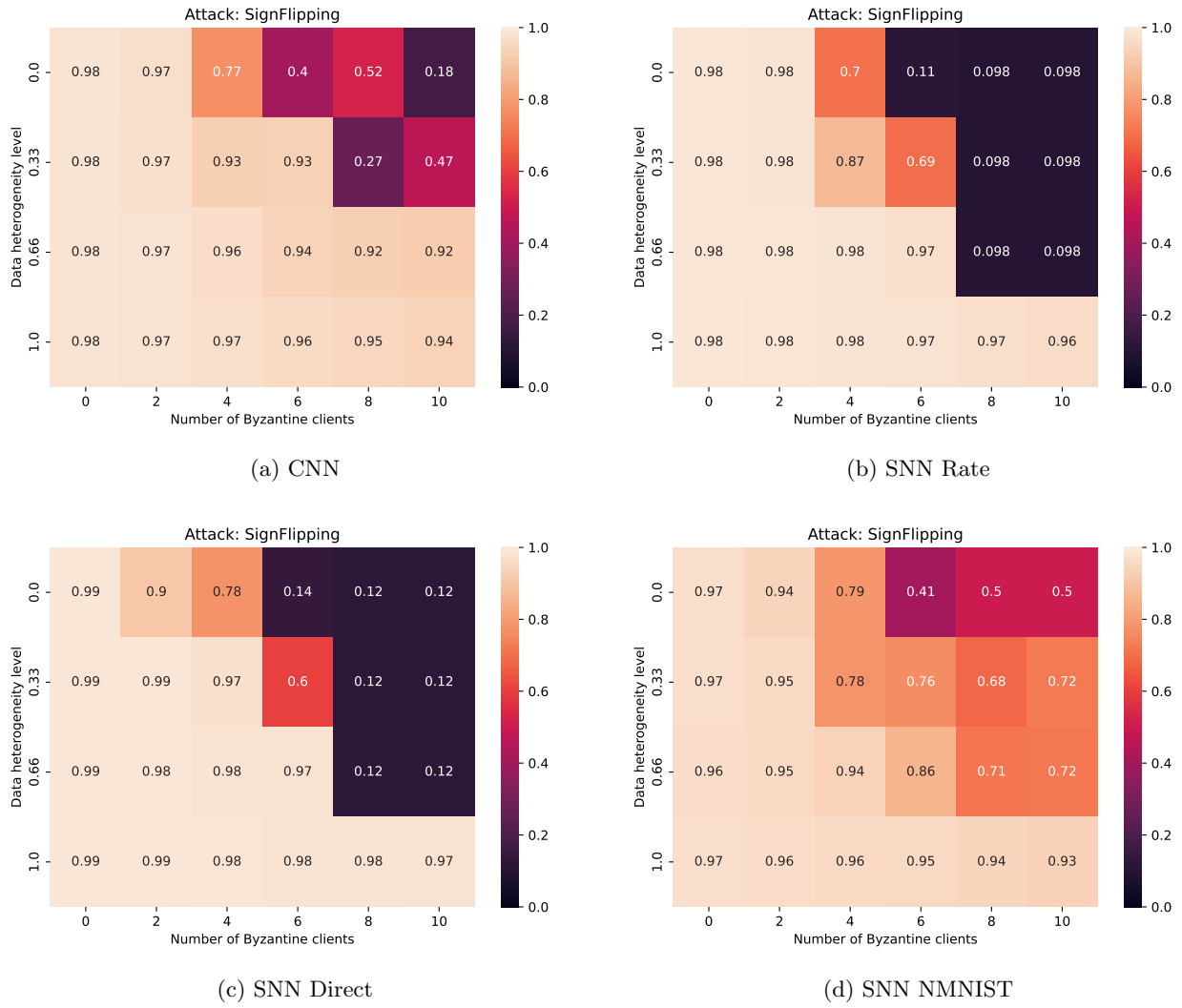


Figure 19: Accuracy comparison under Sign Flipping attack using Geometric Median.

5.4 Multi-Krum

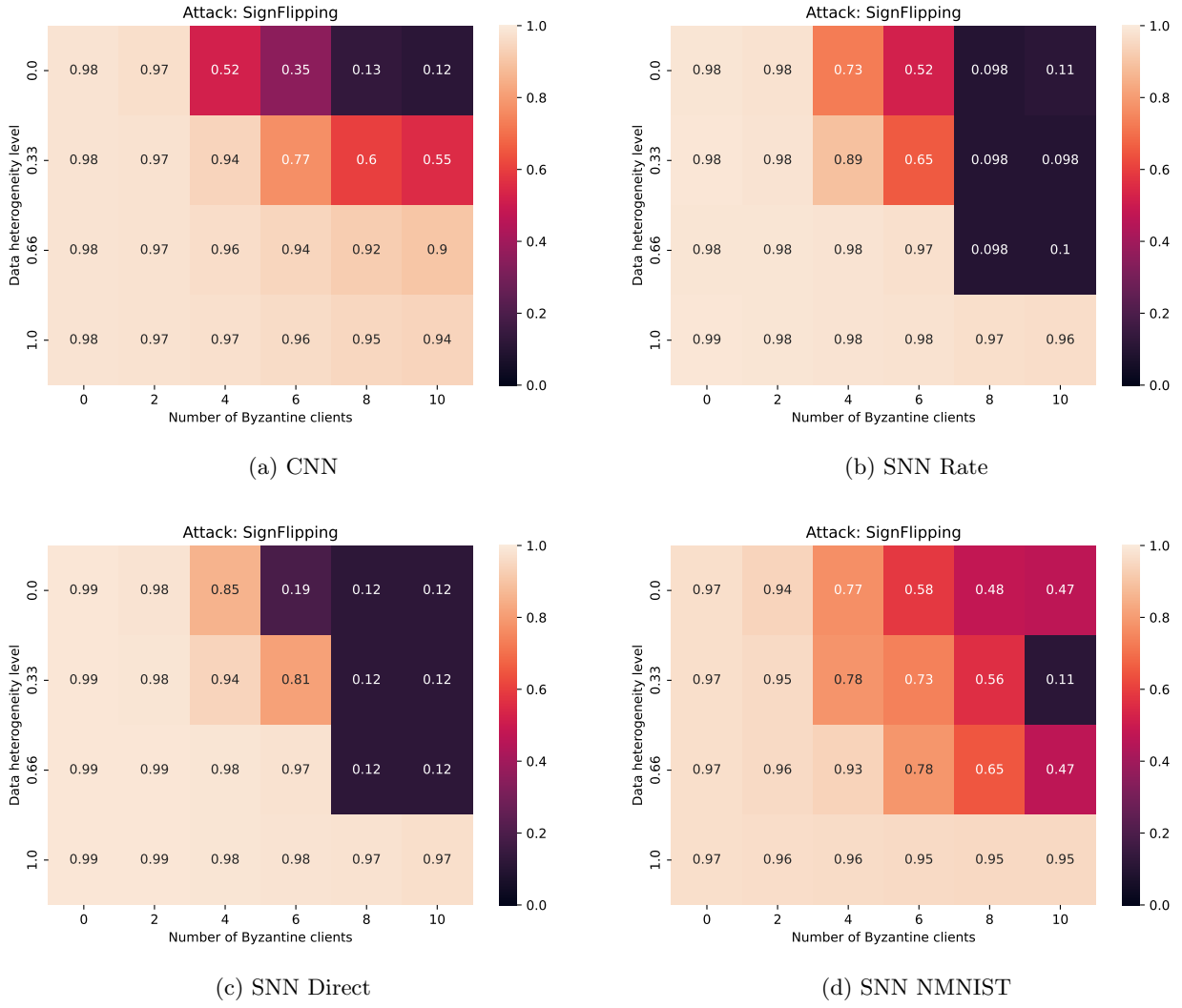
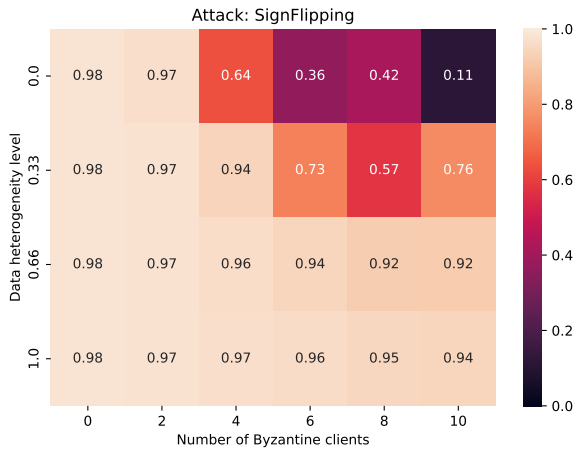
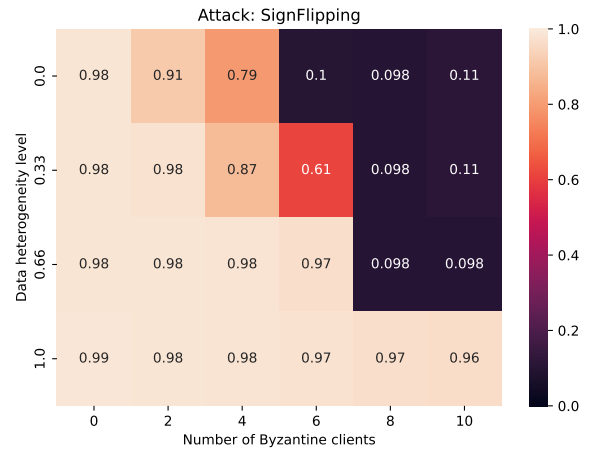


Figure 20: Accuracy comparison under Sign Flipping attack using Multi-Krum.

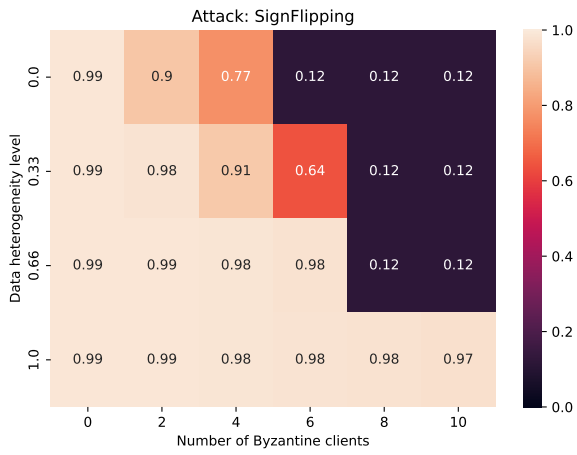
5.5 Trimmed Mean



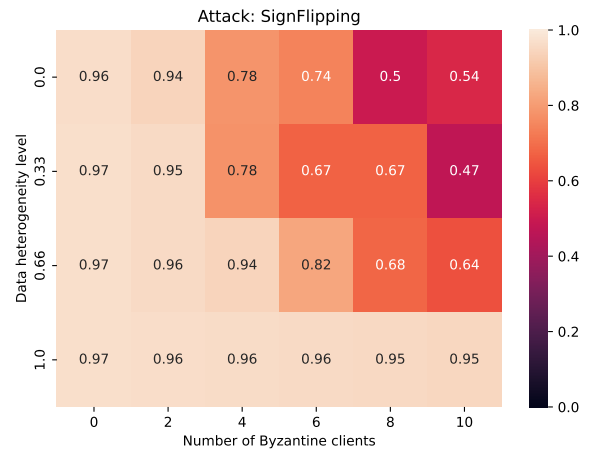
(a) CNN



(b) SNN Rate



(c) SNN Direct



(d) SNN NMNIST

Figure 21: Accuracy comparison under Sign Flipping attack using Trimmed Mean.